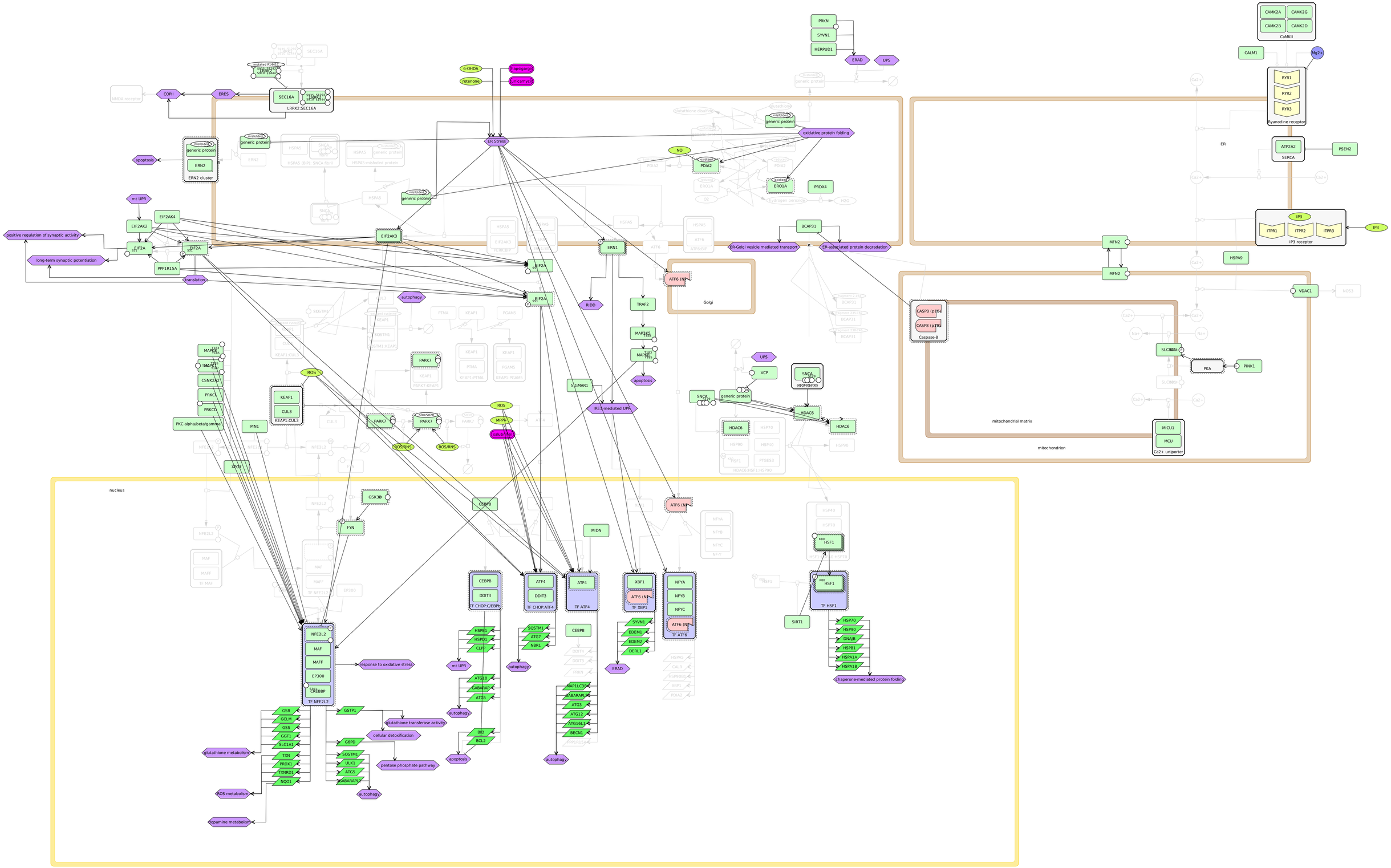

CAMK2A
CAMK2G
CAMK2B
CAMK2D
CaMKII
CALM1
Mg2+
RYR1
RYR2
RYR3
Ryanodine receptor
ATP2A2
SERCA
PSEN2
IP3
ITPR1
ITPR2
ITPR3
IP3 receptor
ER
Ca2+
MFN2
HSPA9
VDAC1
NOS3
CASP8 (p18)
CASP8 (p10)
Caspase-8
SLC8B1
S258
PKA
PINK1
Na+
MICU1
MCU
Ca2+ uniporter
mitochondrial matrix
mitochondrion
PRKN
SYVN1
HERPUD1
ERAD
UPS
misfolded
generic protein
glutathione disulfide
glutathione
unfolded
oxidative protein folding
NO
S-nitrosylation
oxidized
reduced
PDIA2
ERO1A
PRDX4
O2
hydrogen peroxide
H2O
BCAP31
ER-Golgi vesicle mediated transport
ER-associated protein degradation
fragment 2-164
fragment 235-167
fragment 238-246
Golgi
ATF6 (N)
6-OHDA
rotenone
thapsigargin
tunicamycin
ER Stress
HSPA5
PERK:BIP
EIF2AK3
IRE1:BIP
ERN1
ATF6
ATF6:BIP
RIDD
TRAF2
MAP3K5
T838
MAPK8
T183
Y185
apoptosis
SIGMAR1
IRE1-mediated UPR
SNCA
S129
VCP
Ub
aggregates
HDAC6
HSP70
HSP90
HSP40
HSF1
K80
PTGES3
HDAC6:HSF1:HSP90
mutated R1441C
LRRK2
S935
S1292
S910
S1444
SEC16A
LRRK2:SEC16A
ERES
COPII
NMDA receptor
ERN2
ERN2 cluster
HSPA5 (BiP): SNCA fibril
fibril
HSPA5:misfoded protein
mt UPR
EIF2AK4
EIF2AK2
EIF2A
S51
P
PPP1R15A
positive regulation of synaptic activity
long-term synaptic potentiation
translation
autophagy
CUL3
SQSTM1
oxidized cysteine
KEAP1
KEAP1:CUL3
SQSTM1:KEAP1
PTMA
PGAM5
KEAP1:PTMA
KEAP1:PGAM5
PARK7
PARK7:KEAP1
SOH/SO2H
SO3H
ROS
MPP+
salubrinal
ROS/RNS
MAPK8
MAPK1
T185
Y187
CSNK2A2
PRKCI
PRKCD
PKC alpha/beta/gamma
PIN1
NFE2L2
XPO1
FYN
nucleus
GSK3B
MAF
MAFF
TF MAF
EP300
CREBBP
S301
TF NFE2L2
response to oxidative stress
CEBPB
DDIT3
TF CHOP:C/EBPb
ATF4
TF CHOP:ATF4
TF ATF4
MIDN
XBP1
TF XBP1
NFYA
NFYB
NFYC
NF-Y
TF ATF6
HSF1:HSP40:HSP70
SIRT1
TF HSF1
GSR
GCLM
GSS
GGT1
SLC1A1
glutathione metabolism
TXN
PRDX1
TXNRD1
NQO1
ROS metabolism
dopamine metabolism
GSTP1
glutathione transferase activity
cellular detoxification
G6PD
pentose phosphate pathway
SQSTM1
ULK1
ATG5
GABARAPL1
autophagy
HSPE1
HSPD1
CLPP
mt UPR
ATG10
GABARAP
ATG5
autophagy
BID
BCL2
apoptosis
SQSTM1
ATG7
NBR1
autophagy
DDIT4
DDIT3
PRKN
MAP1LC3B
GABARAPL2
ATG3
ATG12
ATG16L1
BECN1
PPP1R15A
autophagy
SYVN1
EDEM1
EDEM2
DERL1
ERAD
HSPA5
CALR
HSP90B1
XBP1
PDIA2
HSP70
HSP90
DNAJB
HSPB1
HSPA1A
HSPA1B
chaperone-mediated protein folding

VDAC1
PINK1
MFN2
IP3
mutated R1441C
LRRK2
S935 S1292
S910 S1444
Mg2+
PSEN2
CALM1
ROS/RNS
HDAC6
PARK7
VCP
HERPUD1
PRKN
SYVN1
PARK7
HSPA9
CAMK2A
CAMK2G
CAMK2B
CAMK2D
CaMKII
autophagy
UPS
PKA
MFN2
IP3
ITPR1
ITPR2
ITPR3
IP3 receptor
PRDX4
oxidative protein folding
NO
BCAP31
SEC16A
LRRK2
LRRK2:SEC16A
RYR1
RYR2
RYR3
Ryanodine receptor
misfolded
generic protein
ERN2
ERN2 cluster
ATP2A2
SERCA
SOH/SO2H
PARK7
generic protein
SNCA
S129
aggregates
SNCA
S129
ERAD
ATF6 (N)
SLC8B1
P
MICU1
MCU
Ca2+ uniporter
CASP8 (p18)
CASP8 (p10)
Caspase-8
rotenone
6-OHDA
thapsigargin
tunicamycin
misfolded
generic protein
oxidized
ERO1A
oxidized
PDIA2
ER-associated protein degradation
ER-Golgi vesicle mediated transport
ERES
ER
HDAC6
ATF4
mt UPR
ER Stress
COPII
EIF2AK4
EIF2AK2
PPP1R15A
EIF2AK3
P
ERN1
ATF6 (N)
EIF2A
S51
EIF2A
S51
P
RIDD
SIGMAR1
TRAF2
nucleus
translation
IRE1-mediated UPR
MAP3K5
T838
ROS
MPP+
PIN1
long-term synaptic potentiation
salubrinal
T185
Y187
MAPK1
PKC alpha/beta/gamma
CSNK2A1
positive regulation of synaptic activity
PRKCI
XPO1
PRKCD
MIDN
K80
HSF1
TF HSF1
ATF6 (N)
XBP1
TF XBP1
GSK3B
CEBPB
NFYA
NFYB
NFYC
ATF6 (N)
TF ATF6
SIRT1
T183
Y185
MAPK8
KEAP1
CUL3
KEAP1:CUL3
ATF4
DDIT3
TF CHOP:ATF4
TF ATF4
MAF
MAFF
EP300
S301
CREBBP
TF NFE2L2
HSP70
DNAJB
HSP90
HSPA1A
HSPA1B
HSPB1
CEBPB
DDIT3
TF CHOP:C/EBPb
EDEM2
EDEM1
SYVN1
DERL1
P
FYN
K80
HSF1
apoptosis
ATG7
NBR1
SQSTM1
MAP1LC3B
ATG3
ATG16L1
ATG12
GABARAPL2
BECN1
ULK1
GABARAPL1
GSTP1
GSR
GSS
SLC1A1
GCLM
GGT1
G6PD
NQO1
response to oxidative stress
ATG5
PRDX1
TXNRD1
TXN
chaperone-mediated protein folding
ATG10
GABARAP
BID
BCL2
HSPE1
HSPD1
CLPP
ERAD
autophagy
cellular detoxification
glutathione transferase activity
glutathione metabolism
pentose phosphate pathway
quinone metabolism
ROS metabolism
apoptosis
mt UPR
P
NFE2L2